

Gated DeltaNet-2: Decoupling Erase and Write in Linear Attention

Ali Hatamizadeh • Yejin Choi • Jan Kautz

arXiv:2605.22791v1 [cs.AI]
21 May 2026



ONE-LINE TAKEAWAY: Gated DeltaNet-2 achieves the strongest overall results among Mamba-2, Gated DeltaNet, KDA, and Mamba-3 variants across language modeling, commonsense reasoning, and retrieval by decoupling erase and write gates in linear attention.



PROBLEM

- Linear attention uses a fixed-size recurrent state that cannot grow with context length, leading to interference between compressed associations.
- Existing delta-rule models tie erasing and writing through a single scalar gate, limiting control over memory editing.



METHOD

- Introduces Gated DeltaNet-2 with channel-wise erase gate b_t and channel-wise write gate w_t to decouple erase and write operations.
- Absorbs cumulative channel-wise decay into asymmetric erase factors using a WY algorithm with chunkwise structure.
- Uses a gate-aware backward pass to preserve efficient parallel training.



RESULTS

- At 1.3B parameters trained on 100B FineWeb-Edu tokens, Gated DeltaNet-2 achieves the strongest overall results among Mamba-2, Gated DeltaNet, KDA, and Mamba-3 variants.
- It improves the evaluated multi-key retrieval setting on long-context RULER needle-in-a-haystack benchmarks.
- It remains strong in both recurrent and hybrid settings for real-world recall.

KEY NUMBERS

1.3B

Number of model parameters

100B

Number of tokens used for training (FineWeb-Edu)

RULER

needle-in-a-haystack benchmarks

Key evaluation task where Gated DeltaNet-2 shows improved performance

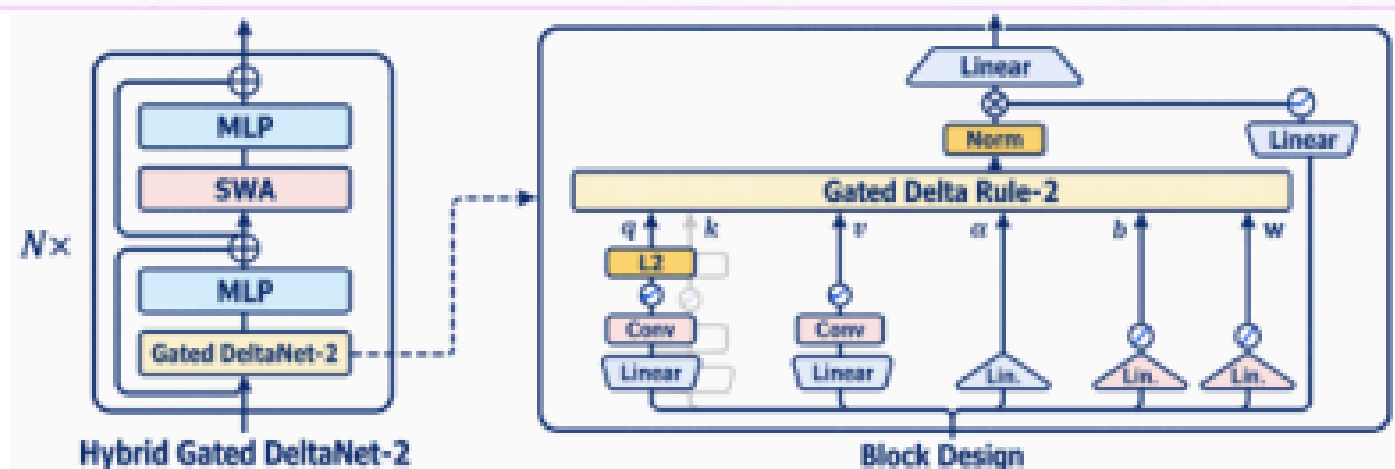
1

The decay clears the state uniformly, while the delta rule edits a selected asso

The decay clears the state uniformly, while the delta rule edits a selected association. This is a useful division of labor, but both gates are scalar per head.

2

Figure 1 | Visualization of the hybrid architecture and block design of Gated De



3

Figure 2 | Training throughput on a H100 GPU.

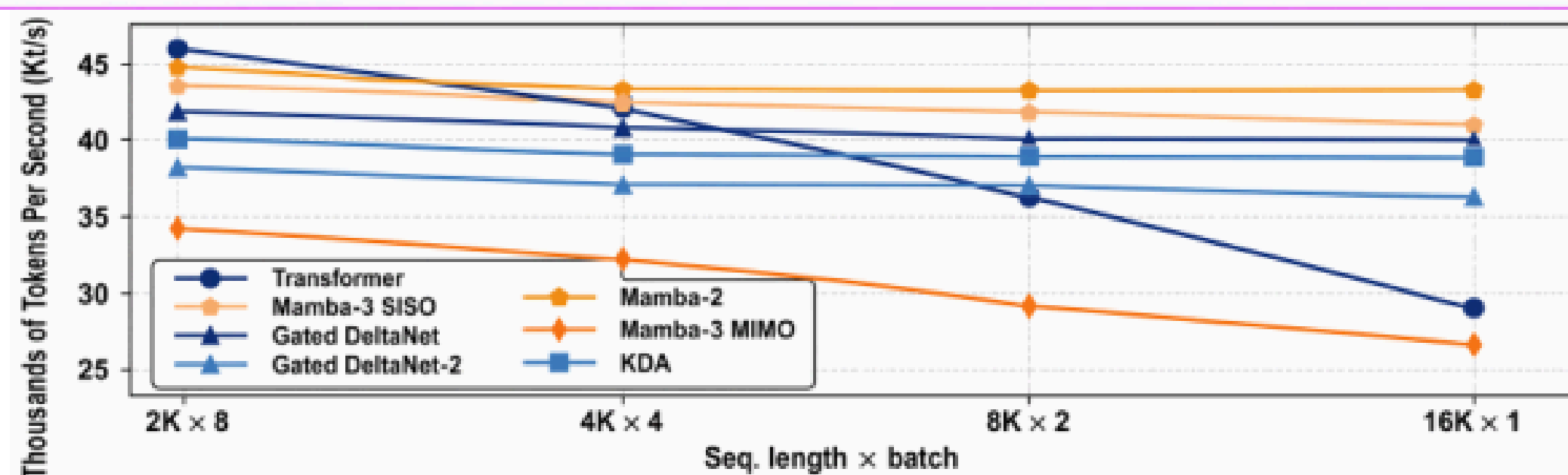


Figure 2 | Training throughput on a H100 GPU.

POSTER SPECIFICATIONS



NeurIPS 2025 (Main Conference)
Landscape
1219 mm x 914 mm
(Push pins on provided boards)



NeurIPS 2025 Workshops
& Mexico City Posters
Portrait
24 in x 36 in
(Lightweight paper)